

01 SCIENCE.md - can != may / Action OS science

Source: gate/SCIENCE.md

Science + tech -> Gate mouth (PRI)

Motive: Contribute massively to Tier-S continuity **with** global coordinators.
Collaborate as the clearance mouth. **Do not own** C2, nukes, grid, or link monopolies.

Formula: Inhibit unjustified irreversible acts (PRI) - fail-closed - as a **driver-node mouth** on the act graph, not a hub dashboard.

Tech thesis: Frontiers multiply what systems **can** do. Gate owns the mouth between **can** and **may** - contribute under each stack; do not own their guns.

First weld (named)

Withdraw / payout - clear before wire.
Prove DENY on one irreversible money-leave path (`POST /v1/payouts/{id}/release`).
Checkout: `/operator` -> withdraw.
Next after prove: grid shed/reconnect (not first). Nuclear C2 stays state-only.

Science -> law

Science	Gate law
PRI (pre-irreversibility inhibition)	`execute` iff LIVE AND justified; else DENY; no bypass; CHARGE-on ly re-open
Fail-closed	uncertainty ? DENY; soft-yes under panic is the failure mode
Thermo intervention (no perfect `do(*)`)	measure -> control -> stranger-verifiable receipt; silent fre e yes illegal
Driver nodes (Liu-Slotine-Barabási)	weld irreversible `*edges*`; hubs != drivers
Quorum / unfireable mouth	LIVE needs policy quorum + CHARGE + Gate; evidence outside actor trust bound ary

Tech frontiers -> mouth

Frontier	Gate mouth	Own?
AI agent control / MCP firewalls	LIVE/DENY on irreversible agent writes; soft prompt != DENY	No - under operator tool graphs
Grid-forming inverters + smart breakers	PRI on shed/reconnect edges after payout prove	No - under utilities
Military LEO mesh / SDN class	Optional fail-closed on irreversible link/priority/release	No - under programs
TEE + MPC + HSM (PQ custody)	Custody proves keys; Gate proves LIVE	No - beside vaults
Planetary-defense kinetic GNC	Conceptual PRI on unjustified commit	No - state owns the shot

Tier S - contribute ladder

1. Commercial driver weld (**payout clear**) - prove DENY
2. Critical infra mouth (grid shed / wholesale finality)
3. Named in coordinator COOP ("clears via Gate") + cleared custodians
4. Defense/comms **release mouth** under programs - `force_production_weld` false until real

Nuclear football: **state only**. We may someday be a tiny component under them - never the sovereign.

Distribution

- Sell PRI to **coordinators**, not consumers
- Land: `/operator` (one door)
- **First weld = payout clear**; grid is next
- Find **driver edges**, not vanity hubs
- Price **authority dissipation** (weld + bps)
- Contracts/copy: collaborate; mouth only; favor to civilization
- Ads floor: `/privacy` + `/terms`; `their_production` false; pixels off by default

Machine-readable

`GET /.well-known/science-pri.json` * `GET /science` * `GET /privacy` * `GET /terms` * `GET /.well-known/legal.json``